PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol
AUTHORS	Kieu, Alexander; Govender, Romona; Östlundh, Linda; King , Jeffrey

VERSION 1 – REVIEW

REVIEWER	Reynolds, Andrew University of Otago, Department of Medicine
REVIEW RETURNED	15-Mar-2021

GENERAL COMMENTS	Thank you for this opportunity to consider this systematic review proposal and blood glucose monitoring regimens. I have only one major comment, and that is if looking to compare F/CGM with SMBG/a1c, please then only list outcomes in your PICO that can be measured by both regimens of blood glucose monitoring. For smaller points: • please consider published articles in languages other that English, this does not necessarily rely on formal translation services. • Please consider non-randomized and non-controlled trials as well as RCTs and observational studies. • Please reconsider the statement in strengths and limitations: “Inclusion of non-peer reviewed, grey literature in the review may reduce the strength of evidence.” • Please show SI units of blood control alongside % (A1c) and mg/dL (blood glucose concentration). • Please consider reporting measures of glycaemic variability from eligible studies when available. Thank you again for this opportunity, all the best with undertaking your research.

REVIEWER	Adler, Alma Brigham and Women's Hospital, Division of Global Health Equity
REVIEW RETURNED	09-Jun-2021

GENERAL COMMENTS	Overall comments: 1. I would suggest where ever possible differentiating between studies that focus on type 1, type 2 diabetes or both.

	2. You need to rewrite your objectives as they do not match between abstract and main document, and they do not match your PICO. 3. A systematic review protocol should be written so that other researches can duplicate your results, so you should make some sections clearer. I have provided some examples below. Abstract-Objectives second objective-I am not sure the right question is "who are the primary care patients" I think you are more interested on characterizing them. Your wording in the main text makes more sense. Introduction: I would move the definition of hypoglycemia (line 44-46) to the methods Objectives-your objectives read very differently in the abstract and main text. please make sure that they match. Your objectives state that "findings of improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction reported in the specialty setting with addition of CGM or FGM translate to the primary care setting?" This suggests that you will be comparing care or outcomes at specialized settings to primary care settings, however your population is just at the primary care setting. I would reword your objectives. Right now your comparison (Pico C) also does not really fit your objectives but rewording your objectives would solve that. Methods -why are you limiting your search to February 2021? -I found the section on the search strategy a little confusing. first you say one is being developed, and then you list a pre-search. With this it is difficult to see how the actual search will look like. Is the intent that you are waiting for peer review comments to come back before you finalize the search strategy, or just that it is not complete yet? -Your outcomes of interest section should be clearer. List primary outcomes clearly and then separately what your secondary outcomes are. You may want to consult other protocols to see how they have done this. -Your data synthesis section also needs to be clearer. You may want to refer to a reference such as the Cochrane handbook as a guide. -You say that you will discuss feasibility of data synthesis at the end, after describing your data synthesis. Do you mean meta-analysis? if so, how do you plan on doing it? what method will you use? What will your criteria for "feasibility" be? You need to specify all this ahead of time. -instead of qualitative narrative do you mean narrative synthesis? There are some formatting issues throughout the protocol, these should be taken care of.

	You may want to include a reference for ROBINS I and II, for example https://www.bmj.com/content/355/bmj.i4919

VERSION 1 – AUTHOR RESPONSE

Reviewer Reports:

Reviewer: 1
Dr. Andrew Reynolds, University of Otago

Comments to the Author:
Thank you for this opportunity to consider this systematic review proposal and blood glucose monitoring regimens.

I have only one major comment, and that is if looking to compare F/CGM with SMBG/a1c, please then only list outcomes in your PICO that can be measured by both regimens of blood glucose monitoring.

Dear Dr. Reynolds, this is a very good point. Our primary outcome of haemoglobin A1c, and our secondary outcomes of hypoglycaemia, hospitalizations and change in baseline A1c will be measured in both the F/CGM and usual care/SMBG groups. In our literature review, we did find some studies using both F/CGM and SMBG in the control arm (in order to compare their outcomes of change in TIR/TAR/TBR to intervention group), but only making clinical decisions based on SMBG. We will note that if studies only measure our secondary outcomes of TIR/TAR/TBR in the intervention arm, then we will only be able evaluate change in baseline to study end.

(Page 11, Lines 22-25 in BMJ's pdf copy; Page 9, Lines 205-206 in authors' submitted copy):
For the secondary outcomes of TIR, TAR, and TBR, we will evaluate change from baseline if studies are unable to compare CGM/FGM intervention to control.

For smaller points:
• please consider published articles in languages other that English, this does not necessarily rely on formal translation services.

Thank you for your suggestion. Ideally, we would include relevant published articles written in all languages rather than just English, but unfortunately, we do not have the resources to accomplish that.

• Please consider non-randomized and non-controlled trials as well as RCTs and observational studies.

Thank you for noticing this, which we have incorporated into our protocol

(Page 10, Lines 5-8 BMJ copy; Page 8, Lines 175-176 authors' copy):
Inclusion criteria are randomised and non-randomised controlled trials, noncontrolled trials, or observational studies with patients of any age with type 1 diabetes…

• Please reconsider the statement in strengths and limitations: "Inclusion of non-peer reviewed, grey literature in the review may reduce the strength of evidence."

Your point is well taken; we have deleted this statement.

• Please show SI units of blood control alongside % (A1c) and mg/dL (blood glucose concentration).

Thank you for this suggestion. We have done this in order to allow readers from a wider global readership to understand the values.

(Page 7, Lines 45-47 BMJ copy; Page 5, Lines 123-124 authors' copy):
American Diabetes Association (ADA) guidelines 2019 for a goal A1c in most patients of less than 7% (53 mmol/mol, 154 mg/dL, 8.6 mmol/L).18

(Page 10, Lines 51-52 BMJ copy; Page 8, Line 195 author copy):
HbA1c of ≤7% (≤53 mmol/mol, ≤154 mg/dL, ≤8.6 mmol/L).

• Please consider reporting measures of glycaemic variability from eligible studies when available.

Great suggestion. If glycaemic variability is reported in the eligible studies, we will include this.

(Page 11, Lines 40-48 BMJ copy; Page 9, Lines 213-216 authors' copy):
If data exists, we will report on glycaemic variability, patient-oriented clinically significant outcomes of patient satisfaction and diabetes related complications such as major cardiovascular events, neuropathy, nephropathy, and retinopathy.

Thank you again for this opportunity, all the best with undertaking your research.

Thank you for your well wishes; we also wish you all the best!

Reviewer 2
Overall comments:
1. I would suggest where ever possible differentiating between studies that focus on type 1, type 2 diabetes or both.

Dear Dr. Adler, thank you for this suggestion. We absolutely agree that the type of diabetes in the study population is absolutely central to the interpretation and application of the results (i.e., we cannot generalize results from a study of people living with type 1 diabetes to the care of people living with type 2 diabetes.) We have adjusted our manuscript accordingly.

(Page 10, Lines 12-15 in BMJ's pdf copy; Page 8, Lines 178-179 in authors' submitted copy):
Included studies will be grouped according to the type of diabetes assessed (type 1, type 2, both type 1 and 2, and gestational diabetes).

(Page 15, Lines 40-45 BMJ copy; Page 13, Lines 303-305 authors' copy):
We anticipate these subgroups to include type 1 diabetes, type 2 diabetes without insulin therapy, type 2 diabetes with only long-acting insulin therapy, type 2 diabetes with multiple daily insulin injections and gestational diabetes.

2. You need to rewrite your objectives as they do not match between abstract and main document, and they do not match your PICO.

Dear Dr. Adler, thank you for pointing out this mismatch. The objectives have been aligned between the abstract, main document, and PICO.

(Page 4, Lines 29-38 BMJ copy; Page 2, Lines 54-58 authors' copy):
This systematic review aims to answer the questions: "Does the use of CGM/FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

(Page 9, Lines 6-15 BMJ copy; Page 7, Lines 152-156 authors' copy):
This systematic review aims to answer the following questions: Does the use of CGM FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction? If so, what subgroups of primary care patients with diabetes are most likely to benefit?

3. A systematic review protocol should be written so that other researches can duplicate your results, so you should make some sections clearer. I have provided some examples below.

Abstract-Objectives
second objective-I am not sure the right question is "who are the primary care patients" I think you are more interested on characterizing them. Your wording in the main text makes more sense.

Dear Dr. Adler, thank you for suggesting this. We have taken the wording in the main text and modified it for use in the abstract. I hope it now reads more clearly.

(Page 4, Lines 29-38 BMJ copy; Page 2, Lines 54-58 authors' copy):
This systematic review aims to answer the questions: "Does the use of CGM/FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

Introduction:
I would move the definition of hypoglycemia (line 44-46) to the methods

Dear Dr. Adler, we agree and have moved the definition into the methods.

(Page 11, Lines 8-11 BMJ copy; Page 9, Lines 199-200 authors' copy):
• frequency of hypoglycaemia (≤3.8 mmol/L, or ≤68 mg/dL) and severe hypoglycaemia (≤ 3.0 mmol/L, or ≤54 mg/dL),

Objectives-your objectives read very differently in the abstract and main text. please make sure that they match.

Yes, we agree. As discussed above, we have restated the objectives in the abstract in order to match those in the main text.

(Page 4, Lines 29-38 BMJ copy; Page 2, Lines 54-58 authors' copy):
This systematic review aims to answer the questions: "Does the use of CGM/FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

(Page 9, Lines 6-15 BMJ copy; Page 7, Lines 152-156 authors' copy):

This systematic review aims to answer the following questions: Does the use of CGM FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction? If so, what subgroups of primary care patients with diabetes are most likely to benefit?

Your objectives state that "findings of improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction reported in the specialty setting with addition of CGM or FGM translate to the primary care setting?" This suggests that you will be comparing care or outcomes at specialized settings to primary care settings, however your population is just at the primary care setting. I would reword your objectives. Right now your comparison (Pico C) also does not really fit your objectives but rewording your objectives would solve that.

Excellent point. Our review is limited to the primary care setting, and we have reworded the objectives accordingly.

(Page 9, Lines 6-15 BMJ copy; Page 7, Lines 152-156 authors' copy):
This systematic review aims to answer the following questions: Does the use of CGM FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction? If so, what subgroups of primary care patients with diabetes are most likely to benefit?

Methods
-why are you limiting your search to February 2021?

Dear Dr. Adler, thank you for your query. We meant to state that the search would occur once the protocol was finalized. We will also perform a final search just prior to submitting the systematic review to ensure that no recent publications meeting our inclusion criteria are omitted.

(Page 12, Lines 21-24 BMJ copy; Page 10, Lines 227-228 authors' copy):
All databases will be searched from their inception to February 2021. A full search update will be performed ahead of manuscript submission.

(Page 13, Lines 10-15 BMJ copy, Page 11, Lines 244-246 authors' copy):
A search update in all databases will be conducted before finalizing the manuscript to ensure the inclusion of relevant studies published after the search.

-I found the section on the search strategy a little confusing. first you say one is being developed, and then you list a pre-search. With this it is difficult to see how the actual search will look like. Is the intent that you are waiting for peer review comments to come back before you finalize the search strategy, or just that it is not complete yet?

We thank the reviewer for bringing this inconsistency to our attention. The section has been re-written to clarify the status of the pre-searches and the search dates.

(Page 12, Line 29 – Page 13, Line 15 BMJ copy; Pages 10-11, Lines 230-246 authors' copy):
Search strategy
A comprehensive, systematic search strategy was constructed by a medical librarian (LÖ) and reviewed by the subject specialists (AK, RDG & JK). Search terms were defined based on the PICO (population, intervention, comparator, and outcomes) and the study inclusion and exclusion criteria as outlined above. PubMed and PubMed's MeSH was used in the pre-search phase in December-January 2021 to systematically identify and refine relevant search term variations and search

techniques. The search strategy developed in PubMed will be adopted and applied in all selected databases without any geographical or publication year restrictions. We will use the “Text Word” field (or similar) for the best possible information retrieval. Hand screening of the references lists of the included studies will also be conducted.

The search strategy developed through pre-searches in PubMed can be found in the online supplementary material file 2. A search log with notes, results, and details for the searches in all included databases in addition to a PRISMA 2020 flow-diagram outlining the details from the search and screening process will be appended to the final review. A search update in all databases will be conducted before finalizing the manuscript to ensure the inclusion of relevant studies published after the search.

-Your outcomes of interest section should be clearer. List primary outcomes clearly and then separately what your secondary outcomes are. You may want to consult other protocols to see how they have done this.

Thank you for this suggestion. We have attempted to make this section clearer.

(Page 10, Line 47 – Page 11, Line 20 BMJ copy; Pages 8-9, Lines 193-204 authors’ copy):
Primary outcome:
We have defined the primary outcome as the meeting of the ADA goals for glycaemic control in most patients of a HbA1c of ≤7% (≤53 mmol/mol, ≤154 mg/dL, ≤8.6 mmol/L).

Secondary outcomes:
We have included secondary outcomes of:
• change from baseline HbA1c,
• frequency of hypoglycaemia (≤3.8 mmol/L, or ≤68 mg/dL) and severe hypoglycaemia (≤ 3.0 mmol/L, or ≤54 mg/dL),
• hospitalization,
• time in target range (TIR),
• time above target range (TAR), and
• time below target range (TBR).36

-Your data synthesis section also needs to be clearer. You may want to refer to a reference such as the Cochrane handbook as a guide.
-You say that you will discuss feasibility of data synthesis at the end, after describing your data synthesis. Do you mean meta-analysis? if so, how do you plan on doing it? what method will you use? What will your criteria for "feasibility" be? You need to specify all this ahead of time.
-instead of qualitative narrative do you mean narrative synthesis?

Thank you for pointing out the lack of clarity, which is indeed confusing. We do not intend to perform a meta-analysis at this time, so we have removed that section.

We do indeed mean narrative synthesis and have substituted the correct terminology. Thank you.

(Page 15, Line 31 – Page 16, Line 20 BMJ copy; Pages 13-14, Lines 299-317 authors’ copy):
Data synthesis
In a comprehensive table, we will summarize the PICO characteristics of each study including all relevant interventions and outcomes. Thereafter we will determine which studies have similar characteristics and outcomes, which will allow us to group them and compare their relevant findings.

We anticipate these subgroups to include type 1 diabetes, type 2 diabetes without insulin therapy, type 2 diabetes with only long-acting insulin therapy, type 2 diabetes with multiple daily insulin injections and gestational diabetes. Given our review will be inclusive of many study designs, we may group data according to their respective study design. Quantitative data to be reported for each study will include the mean HbA1c at study end of the intervention group (CGM/FGM) compared to control. We will then take the mean differences with the standard errors of the means, calculate the 95% CIs, and graphically represent our results. Weighted mean differences of change from baseline HbA1c, TIR, TAR, and TBR after CGM/FGM intervention will similarly be analyzed. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be reported with their respective 95% CIs, along with qualitative data on patient and physician satisfaction. If any deviation from protocol plans is necessary, modifications to planned comparisons between studies will be made and noted. A narrative synthesis will be formulated for each subgroup of persons with diabetes, and we will give an appraisal of the utility of the addition of CGM or FGM, analyzing which subgroups appear to benefit most from CGM/FGM.

There are some formatting issues throughout the protocol, these should be taken care of.

We have converted the >= and <= to their appropriate symbols, and have adjusted the spacing of the document.

You may want to include a reference for ROBINS I and II, for example
https://eur03.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.bmj.com%2Fcontent%2F355%2Fbmj.i4919&data=04%7C01%7Cjking%40uaeu.ac.ae%7C823a84b8411b43254a0708d9349c9827%7C97a92b044c8743419b08d8051ef8dce2%7C0%7C0%7C637598669455938413%7CUnknown%7CTWFpbGZsb3d8eyJWIjoiMC4wLjAwMDAiLCJQIjoiV2luMzIiLCJBTiI6Ik1haWwiLCJXVCI6Mn0%3D%7C1000&sdata=tq%2BPS5WjlLlpYiM6SLM3X5IKb3aSzbtCzLqPvuQIcHQ%3D&reserved=0

Thank you. We have decided that we will use NHLBI for the quality assessment, and we have referenced accordingly.

(Page 15, Lines 8-13 BMJ copy; Page 13, Lines 289-291 authors' copy):
Two independent reviewers (JK & RDG) will evaluate the risk of bias and the quality of evidence of all included studies using the National Heart, Lung, and Blood Institute (NHLBI) quality assessment tools.41

VERSION 2 – REVIEW

REVIEWER	Reynolds, Andrew University of Otago, Department of Medicine
REVIEW RETURNED	24-Jul-2021

GENERAL COMMENTS	Thank you for this opportunity to consider this systematic review proposal and blood glucose monitoring regimens. A previous reviewer requested some consistency checking between sections of the document - I remain in the belief that this is still required. For example, lines 55-58 state the objective of the review, but what is the comparison? i.e If you are looking to comment on improved glycaemic control, what are you comparing F/CGM use with? If you are comparing blood glucose variables

	following F/CGM use with no F/CGM use, how will you calculate your outcomes TIR/TBR/TAR with out F/CGM use? There is insufficient clarity for someone to duplicate the processes included in this protocol - please use simple phrasing to clearly state what you will do.

REVIEWER	Adler, Alma Brigham and Women's Hospital, Division of Global Health Equity
REVIEW RETURNED	09-Aug-2021

GENERAL COMMENTS	Thank you for your updates to your protocol. It reads much clearer now. I have just a couple of extra comments: Your abstract has a lot of acronyms in it now, can you please make sure you spell these out. Information sources: I see that you list Pubmed and Medline as databases that you will search. Pubmed is a tool for searching Medline, so if you use Pubmed, there is no reason to further search Medline. Data management-you state that screening and extraction are blinded twice. Data synthesis-You do not specify that you will do meta-analysis, are you planning on doing a meta-analysis? It may be that the heterogeneity is too great, but i would suggest attempting it.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1

Dr. Andrew Reynolds, University of Otago

Comments to the Author:

Thank you for this opportunity to consider this systematic review proposal and blood glucose monitoring regimens.

A previous reviewer requested some consistency checking between sections of the document - I remain in the belief that this is still required. For example, lines 55-58 state the objective of the review, but what is the comparison? i.e If you are looking to comment on improved glycaemic control, what are you comparing F/CGM use with? If you are comparing blood glucose variables following F/CGM use with no F/CGM use, how will you calculate your outcomes TIR/TBR/TAR with out F/CGM use?

There is insufficient clarity for someone to duplicate the processes included in this protocol - please use simple phrasing to clearly state what you will do.

Dear Dr. Reynolds, thank you for pointing out this inconsistency. We have included the comparison in our objective statement (Lines 53-57 and 154-158):

This systematic review aims to answer the questions: “Compared to usual care of self-monitoring blood glucose and HbA1c, does the addition of CGM/FGM use in the primary care of patients with diabetes improve glycaemic control, decrease rates of hypoglycaemia, and improve patient and physician satisfaction?” and if so, “What subgroups of primary care patients with diabetes are most likely to benefit?”

Additionally, we clarified our plans of calculating the outcomes of TIR/TBR/TAR when F/CGM is used in the control arm and not used in the control arm (Lines 207-211):

From our preliminary literature review, some studies used both SMBG and CGM/FGM in the control arm (but making clinical decisions based on SMBG) in order to compare change in TIR, TAR and TBR from baseline to study end to the intervention arm of CGM. However, for studies only evaluating change in TIR, TAR, and TBR in the intervention arm without a control, we will assess the percentage change from baseline to study end in these outcomes.

Reviewer: 2

Dr. Alma Adler, Brigham and Women's Hospital, Harvard Medical School

Comments to the Author:

Thank you for your updates to your protocol. It reads much clearer now. I have just a couple of extra comments:

Your abstract has a lot of acronyms in it now, can you please make sure you spell these out.

Dear Dr. Adler, thank you for bringing this to our attention. We have spelled out most of the acronyms in the abstract (Lines 63-64 and 67-68):

time in range (TIR), time below range (TBR), time above range (TAR)

National Heart, Lung, and Blood Institute (NHLBI) quality assessment tool and Grading of Recommendations Assessment, Development and Evaluation (GRADE)

Information sources: I see that you list Pubmed and Medline as databases that you will search. Pubmed is a tool for searching Medline, so if you use Pubmed, there is no reason to further search Medline.

We thank the reviewer for the relevant observation and suggestion. PubMed and Medline do indeed build on the approximate same index with the exception that PubMed covers more journals from a broader subject area compared to Medline and includes pre-indexed materials.

However, differences in the search results can also be seen between the two databases depending on platform used for the Medline search, search technical possibilities connected to the specific platform, etc.

In our case, where our aim is to capture all possible evidence available for a novel and narrow topic, we have chosen to include both Medline and PubMed to increase the result. By systematically repeating the search in Medline (Clarivate's interface), we receive 31 additional, unique papers (uploaded to and de-deplicated in EndNote) compared to searching in PubMed only. Combining both, therefore, increases the search results for the review.

Data management-you state that screening and extraction are blinded twice.

Thank you for noticing this. We have removed the repetition (Lines 255-259):

Data management

We will upload all records identified in the database search to Covidence systematic review software40 for automatic de-duplication and blinded screening, conflict resolving, study selection, and data extraction. Studies identified to be included in the review will be exported to EndNote, a citation management tool.

Data synthesis-You do not specify that you will do meta-analysis, are you planning on doing a meta-analysis? It may be that the heterogeneity is too great, but i would suggest attempting it.

Thank you for your suggestion. We will plan to attempt a meta-analysis if we are able (Lines 324-330):

If there are sufficient RCTs with low variability among them, a meta-analysis will be attempted comparing CGM/FGM to SMBG. A sensitivity analysis will be conducted by excluding studies with a high risk of bias, which will confirm the robustness of our results. We will assess heterogeneity using the I2 statistic, where ≥50% is considered heterogenous.[42,43] If heterogeneity is determined, a meta-regression analysis will be performed to evaluate whether baseline HbA1c, duration of CGM/FGM use, and study design influence CGM/FGM's effects on HbA1c. Egger's regression test will be used to detect publication bias.[44]

VERSION 3 – REVIEW

REVIEWER	Adler, Alma Brigham and Women's Hospital, Division of Global Health Equity
REVIEW RETURNED	17-Aug-2021

GENERAL COMMENTS	Thank you for your revisions